ARYAN SINGH

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SUMMARY

Computer Science and Mathematics student at UIUC building full stack applications, Linux infrastructure, and AI/ML systems. Projects include an LLM gateway, CUDA inference, and FPGA scheduling research.

EDUCATION

University of Illinois Urbana-Champaign Urbana, IL
B.S., Computer Science and Mathematics | GPA: 3.78 / 4.00 *Expected Graduation May 2028*

- Coursework: Machine Learning, Artificial Intelligence, Parallel Programming (CUDA), Algorithms, Distributed Systems, Computer Architecture, Systems Programming, Compilers, Database Systems
- Honors: 1st of 400, CS 341 Malloc Contest (top-performing memory allocator); 4th place, Advanced Track, UIUC CTF 2026; 3rd place, UIUC CTF 2025; 5th of 50 schools, Robot Tour, UIUC State Science Olympiad 2024

EXPERIENCE

University of Illinois Chicago Chicago, IL
Undergraduate Researcher | Prof. Debjit Pal's research group *Aug 2025 - Present*

- Co-authored BoostedSOSA, a dual-FPGA HPC scheduler replacing unreliable user-provided runtime estimates with ML predictions; paper reports **63.85%** lower error and **17x** speedup.
- Currently implementing two zero-copy, lock-free host-FPGA ring buffers to replace flag polling and reduce PCIe handoff overhead in the continuous scheduling pipeline.

Software Engineering Intern *May 2025 - Aug 2025*

- Converted an FPGA scheduler from batch-oriented I/O into a free-running streaming pipeline so HPC jobs could be scheduled as they arrive instead of waiting for batches to fill.
- Built nonblocking Vitis HLS Reader/Writer kernels and multithreaded C++/XRT host I/O; paper measured **<4 us/job** for streaming vs. **245 ms** batching at 100 jobs/s.
- Designed and verified the systolic scheduling datapath, virtual FIFO, and cost calculator with ~1,100 lines of Monte Carlo-driven Vitis HLS tests on Alveo U55C.

SYSTEMS AND SOFTWARE PROJECTS

Advanced Systems Programming (CS 341) | *C, Linux/UNIX, POSIX Threads, TCP/IP*

- Memory Allocator: Built a C allocator for efficient memory reuse, winning **1st place out of 400 students**; used address-sorted free lists, coalescing, fragmentation-aware splitting, and in-place realloc growth.
- Network Server: Enabled **thousands of concurrent TCP clients** on one thread with a C epoll reactor and length-prefixed binary protocol; per-connection state machines resume partial I/O after EAGAIN without busy-waiting.
- UNIX Shell: Built a POSIX shell in C to run commands and manage UNIX processes, implementing fork/exec, signal handling, I/O redirection, and advanced shell commands.
- Parallel Execution: Parallelized builds and data processing with a multithreaded make clone and local MapReduce; used mutexes and condition variables to coordinate work and prevent deadlocks.

Freeway | *TypeScript, Node.js, SQLite* GitHub

- Built a self-hosted gateway so applications can use **12 LLM providers** through one OpenAI-compatible API, with no external runtime dependencies and providers added through JSON configuration.
- Reduced paid upstream calls by **80% on a 184-request benchmark** with four-tier SQLite caching: exact matches, normalized rewordings, equivalent models, and in-flight request coalescing.
- Kept conversations continuous during provider failures with four-step context refitting; migrated a **41k-token thread to an 8k-context model** and routed requests by EWMA latency.
- Classified failures before retries and reserved quota at dispatch to prevent concurrent requests from exceeding rate limits; validated with **598 tests across 29 CI suites**.

EnvSync | *Rust, Spring Boot, PostgreSQL, Next.js* GitHub

- Built a Rust CLI for secure .env sharing and versioning, used by my ACM team on its projects; added masked diffs, conflict checks, and rollback with preserved snapshot history.
- Implemented client-side **XSalsa20-Poly1305 encryption** with Argon2id key derivation and Keycloak OIDC authentication; added a Next.js dashboard for project management and snapshot history.

Homelab: Self-Hosted Production Infrastructure
Proxmox, Kubernetes (K3s), OPNsense, Cloudflare

- Operate public websites and projects serving **30K+** monthly unique visitors and **1M+** requests per month on a 3-node HA K3s/Proxmox cluster; automate rebuilds with Ansible.
- Secure public services with an OPNsense firewall, Suricata and CrowdSec intrusion detection, Cloudflare mTLS, and Authelia single sign-on; harden Linux hosts with nftables and fail2ban.
- Monitor hosts, containers, Proxmox, and the firewall with **16 Prometheus scrape jobs and 26 alerting rules**, plus Grafana/Loki and GeoIP-enriched dashboards that surface malicious traffic in real time.
- Automate builds, tests, and K3s deployments on push with Jenkins CI/CD and the GitHub Checks API.

GPT-2 Inference Engine from Scratch gpt2.aryan-singh.dev
CUDA C++, Nsight, CUTLASS | 4-person team

- Accelerated GPT-2 text generation on an NVIDIA A40 GPU, reducing per-token latency from **311 ms to 12.6 ms** through KV caching over a preallocated device cache.
- Implemented the FlashAttention-2 forward kernel, tiling Q/K/V through shared memory and registers so the full attention probability matrix never reaches GPU DRAM.
- Removed a 32-way shared-memory bank conflict with an XOR swizzle (**1.37x kernel speedup**) and profiled tile shapes; the team's stack reached **24.2x end-to-end speedup** (249.8 ms to 10.3 ms).
- Used an Agile workflow to track tasks and development progress, keeping the 4-person team on track.

Technical Skills

Languages: C, C++, Python, Rust, Java, TypeScript, SQL, Bash
Systems and Networking: Linux/UNIX, POSIX, concurrency, TCP/IP, memory management
Backend and Security: Node.js, Spring Boot, REST APIs, PostgreSQL, SQLite, OIDC/OAuth2, mTLS
Cloud and DevOps: Docker, Kubernetes (K3s), Ansible, Jenkins CI/CD, Prometheus, Grafana
Performance and Accelerators: CUDA, Vitis HLS, XRT, Nsight, GDB, profiling
Development Practices: Git, Agile workflow

Course Certificates

Anthropic Academy: AI Fluency (Framework and Foundations); Introduction to MCP
Hugging Face: AI Agents Course; LLM Course
Cloud Fundamentals: AWS; Microsoft Azure